

Annual Review for SOGNO

SOGNO

Brief Description:

SOGNO creates plug-and-play, cloud-native, micro-services to implement next generation of data-driven monitoring and control systems for grid automation. It simplify the life of utilities by enabling them to optimize their network operations through open source to deliver cost-effectively, and seamless, secure power supply for their customers. SOGNO introduces the idea of grid automation as a modular system in which components can be added through time. This is in opposition to classical monolithic solutions.

TSC Chairperson:

Antonello Monti (amonti@eonerc.rwth-aachen.de)

TSC Members and Affiliations:

Juan Adolfo Galeano, Florian Oppermann, Sanket Gaikwad, Sreejith Pananchickal, Leonardo Carreras *(RWTH Aachen University)*
Varshitha Ramanna, Pranav Kulkarni, Beyza Cizmeci, Varshitha Chamanahalli *(Fraunhofer FIT)*
Markus Mirz (PSI), Roberto Raffo (Google*),* Alberto Patrizi (ARETI)
Holger Blasum (UrStrom Bürgerenergie eG)

Key Links:

GitHub: https://github.com/sogno-platform

Website:

https://www.lfenergy.org/projects/sogno/

Artwork: *N/A*

Mailing lists:

https://lists.lfenergy.org/g/SOGNO-TSC

OpenSSF Best Practice Badge URL:

https://www.bestpractices.dev/en/projects/5627

Organizations/projects contributing/using SOGNO

































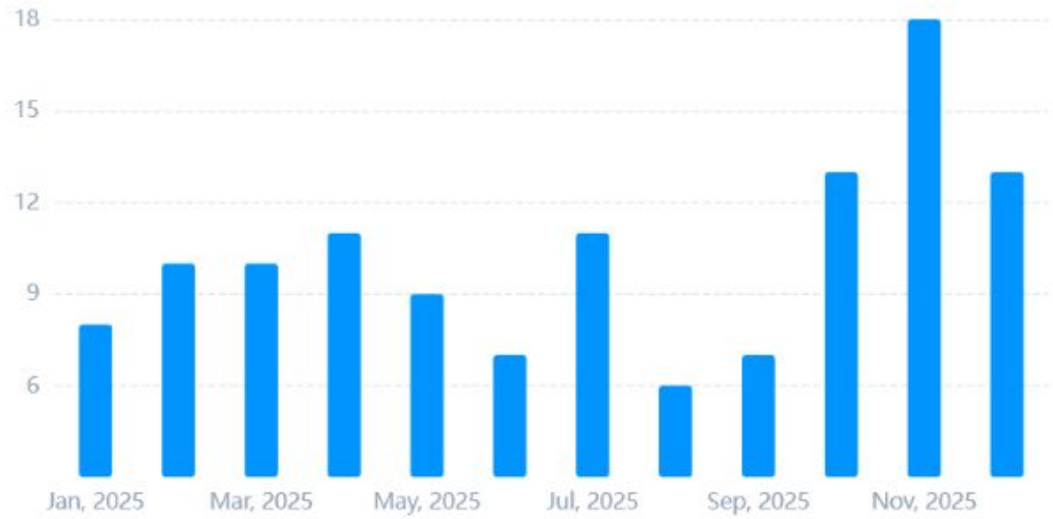
Contributions:

40 ↑ 25% (+8) vs. 32 last period

Maintainers 8

Reviewers 13

Weekly | Monthly | Quarterly



Organizations leaderboard

Organizations ranked by the number of contribution activities performed by contributors on their behalf during the selected time period. Learn more

Include collaborations

All activities

Organization	Total contributions
RWTH Aachen University	1,587 - 49%
Tom's Café	809 - 25%
OPAL-RT Germany GmbH	481 - 15%
Universidad Nacional de Colombia	153 - 5%
E.ON Energy Research Center	125 - 4%
Loophole Labs Inc	21 - 1%
OZ Sports	17 - 1%
Zaphiro Technologies	10 - 0%
FGH	8 - 0%
Swissgrid AG	5 - 0%

Contributions:

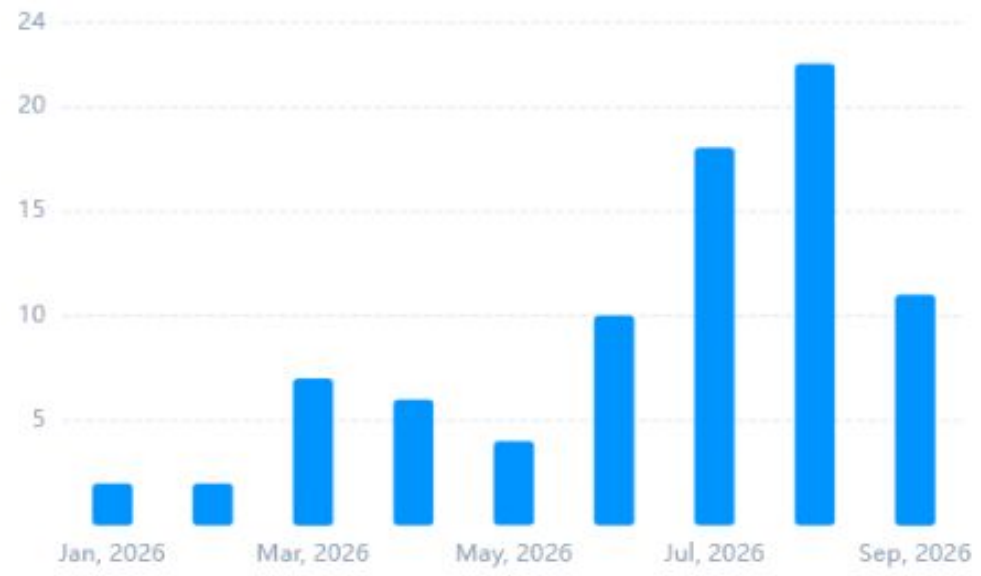
Active contributors

34 3% (+1) vs. 33 last period

Maintainers 0

Reviewers 9

monthly active contributors



Organizations leaderboard

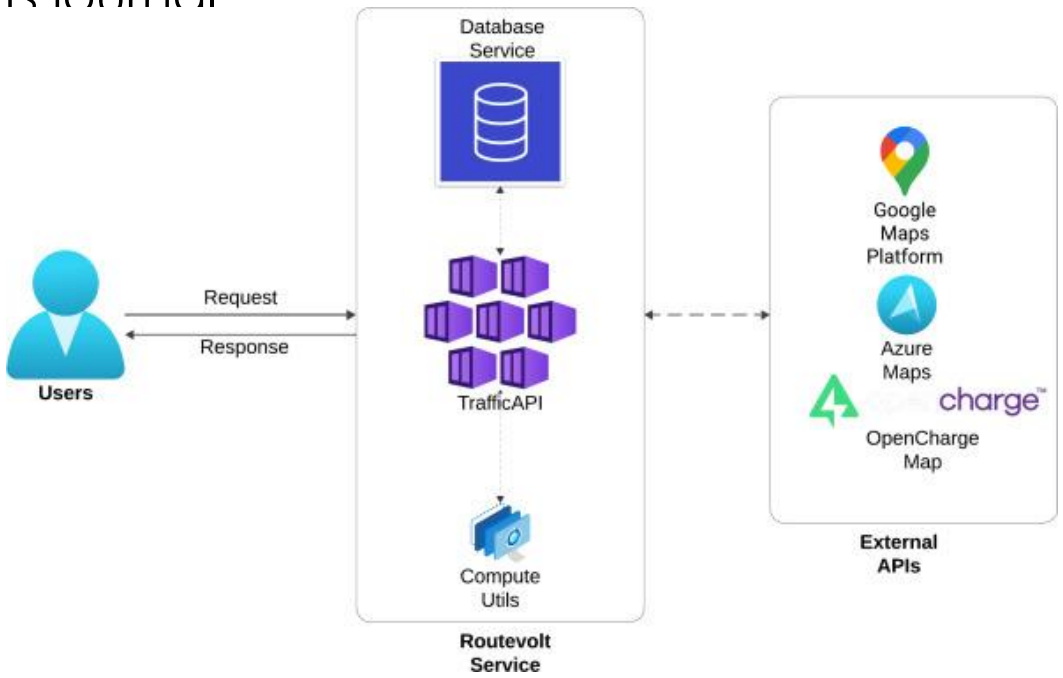
All activities

Organization	Total contributions
RWTH Aachen University	3,015 - 44%
E.ON Energy Research Center	2,030 - 29%
Tom's Café	765 - 11%
OPAL-RT Germany GmbH	430 - 6%
PSI Software AG	379 - 5%
Ita Institut Fur Textiltechnik Der Rwth Aachen Uni	120 - 2%
OPAL-RT TECHNOLOGIES	52 - 1%
Universidad Nacional de Colombia	36 - 1%
FGH	22 - 0%
OZ Sports	17 - 0%

Key Achievements in the past year

Routevolt: New Service

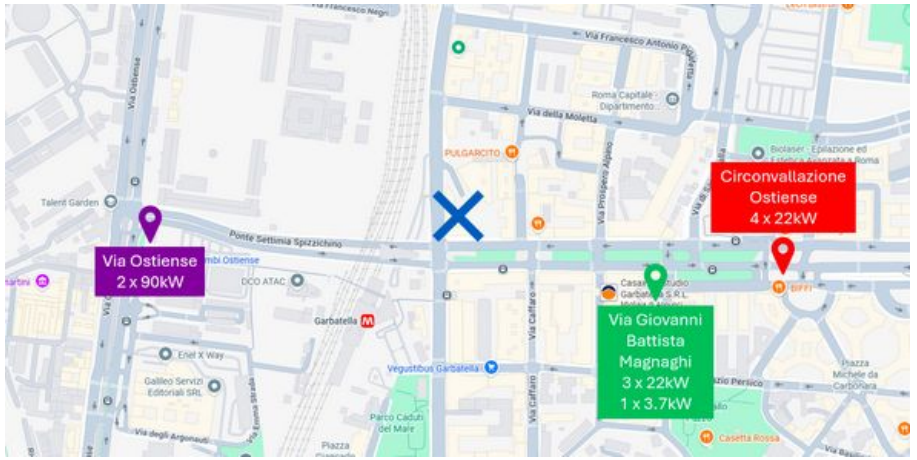
- Is being used as real-time traffic endpoint for evrich. (Advanced estimation of arrival SOC and time calculations to charging points)
- Software paper published in Elsevier Software Impacts journal
- Now available in SOGNO repo.
- DOI: https://doi.org/10.1016/j.simpa.2026.100837

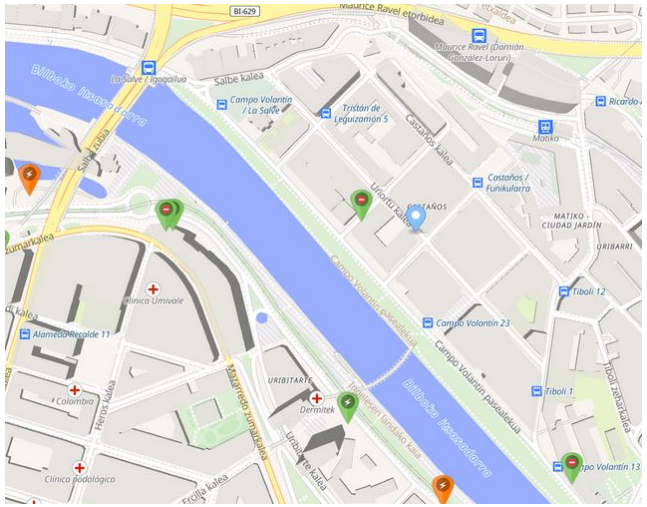


Key Achievements in the past year

Evrich & routevolt: Conducted two demos within two EU projects

- **FLOW: Rome demo**
  - Management of 3 charging stations with total 7 charging ports, helping assess the impact of optimized station selection on reducing charging-related loads by comparing baseline and integrated consumption levels with ARETI.
- **BeFlexible: Bilbao demo**
  - Optimized CP recommendations throughout the 24 h, aiming to improve energy distribution and reduce congestion through a before-and-after analysis of demand patterns with i-de.





Key Achievements in the past year

Pyvolt and PMU based observability demonstrated in the field

BeFlexible: Rome demo

The pyvolt was used with a low cost pmu devoloped by RWTH Aachen. This was used in a secondary substation in Rome with ARETI. The data streamed was used for state estimation, improving the observability and congestion detection.





200 kW Max nominal load

66kW - Average load

Key Achievements in the past year

Proloaf Upgrade



- Improved handling of time features
- Preprocessing steps to create generalizable model (needs evaluation)
- Various QOL improvements
- Integration as Event based Service in OpenEnergyTwin
- Parallel Training now possible

PyMFM Upgrade

- Handling of life battery data
- Various QOL improvements
- Major code restructuring (under review)

Key Achievements in the past year

Evrich Upgrade

- Helm Version (unreleased)
- Improved Simulation allowing Experiment chains and comparison of scenarios (unreleased)

Pipeline Prototype

- Prototype for eventbased architecture.
- Based on Kafka
- Forecasting integrated
- Docker, K8s, Helm
- OIDC under review

Key Achievements in the past year



DPsim

Projects: SEGURO, HYPOWER, CRETE VALLEY

- **New Release: v1.2.1 on Dec 2025**

Available for Linux in PyPI and from source in Github

- **Simulation Models**

State-Space-Nodal component models added, Ph1 and Ph3 line models for Diakoptics

- **Real-Time & VILLAS Enhancements**

Decoupling line EMT models for co-simulation

- **Workflow & CI/CD Upgrades**

Improvements in PyPI publishing, addition of nix build and optimization of container builds.

- **Testing & Examples**

Easy access for user tests with myBinder including VILLASnode support (badge in the landing page of the repository), example Notebooks updated for new models.

- **Code Quality & Maintenance**

Coverage testing added, workflow tests with pre-commit, documentation updates.

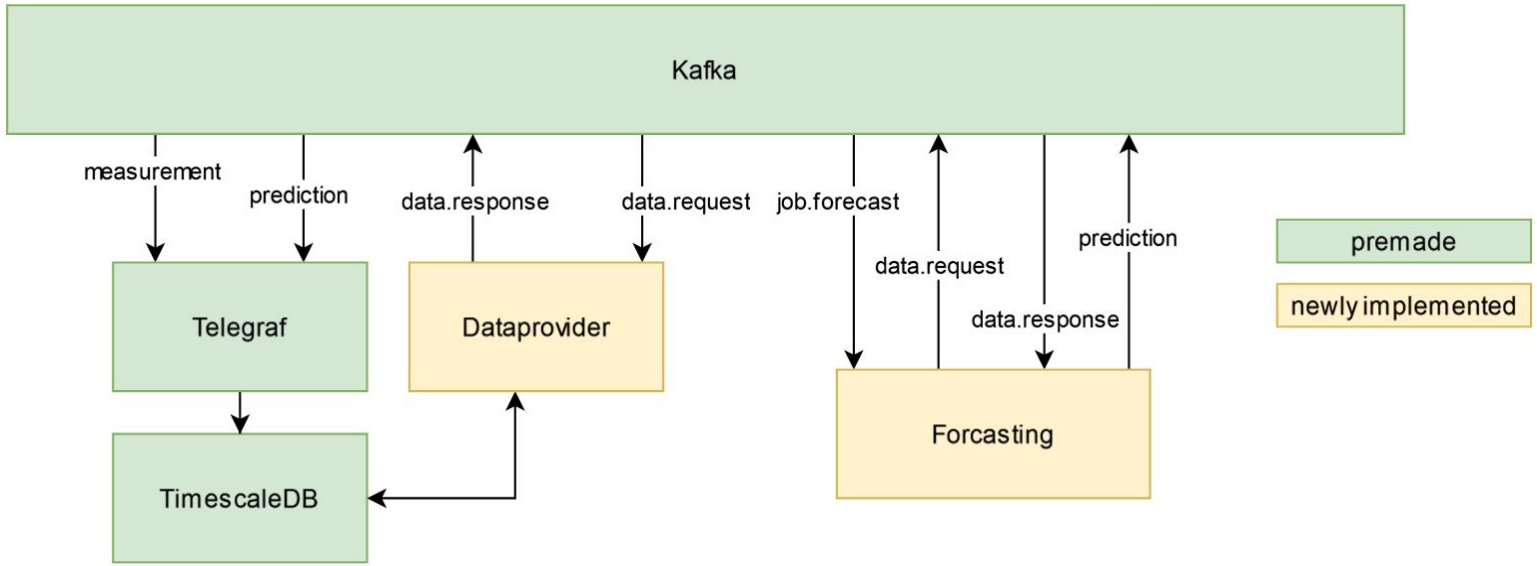
- **Security Aspects**

Trusted publisher added to PyPI and TestPyPI. Progressive dependence upgrades.

Key Achievements in the past year

Continuing adapting the data Architecture (Prototype)

- Prototype
  - Unified way of data and task exchange
  - Expect to tie disparate SOGNO services together
- Workers are always available
- Central data storage
- Both data and tasks come via Kafka
- Services request missing data,
- Providers fulfil requests autonomously



Prototype architecture and preliminary topics

Key Achievements in the past year

- **InterSCADA** (EU project)
  - Develop an open-source software framework for modular SCADA platform.
    - Checking orchestration mechanism to onboard new microservices. Discussing licensing. There are currently 9 functional microservices.
  - Help grid operators maintain system stability in increasingly hybrid AC/DC power grids
- **OpenEnergyTwin** (OFFIS, RWTH, and Fraunhofer):
  - Integrated Proloaf into OET platform.
  - Advance GUI for visualization of high and low voltage scenario.
  - Project Phase 1 successfully delivered.
  - Preparation for Phase 2.
  - OET results to be used to enhance SOGNO core

Key Achievements in the past year

OpenEnergyTwin (OET) – Phase 1

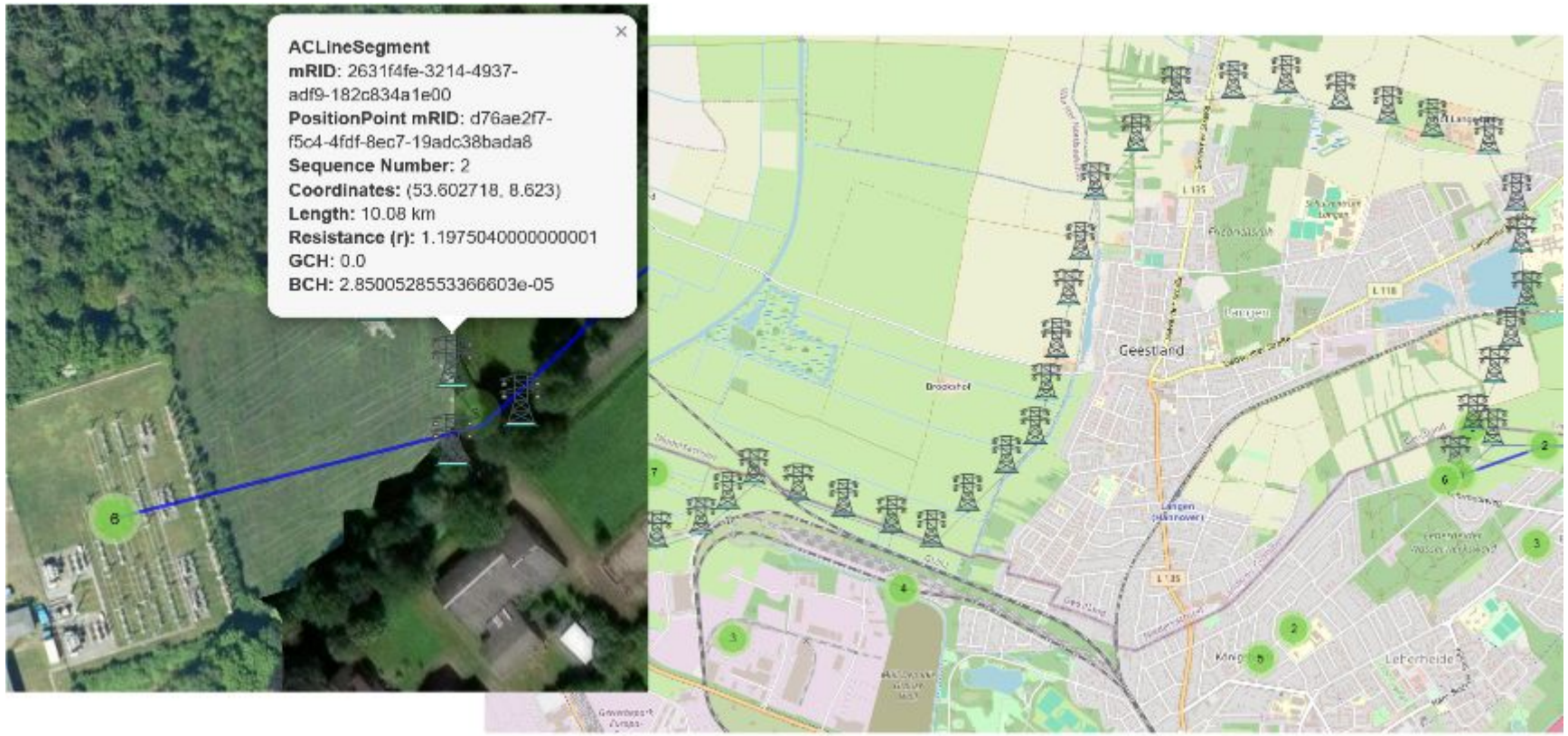
- The human-machine interface auto-drawing topological data and visualizing key live parameter



Key Achievements in the past year

OpenEnergyTwin (OET) – Phase 1

- The human-machine interface auto-drawing topological data and visualizing key live parameter



Key Achievements in the past year

Workshops in ACS and FIT

- A target picture for modular control systems development in Aachen

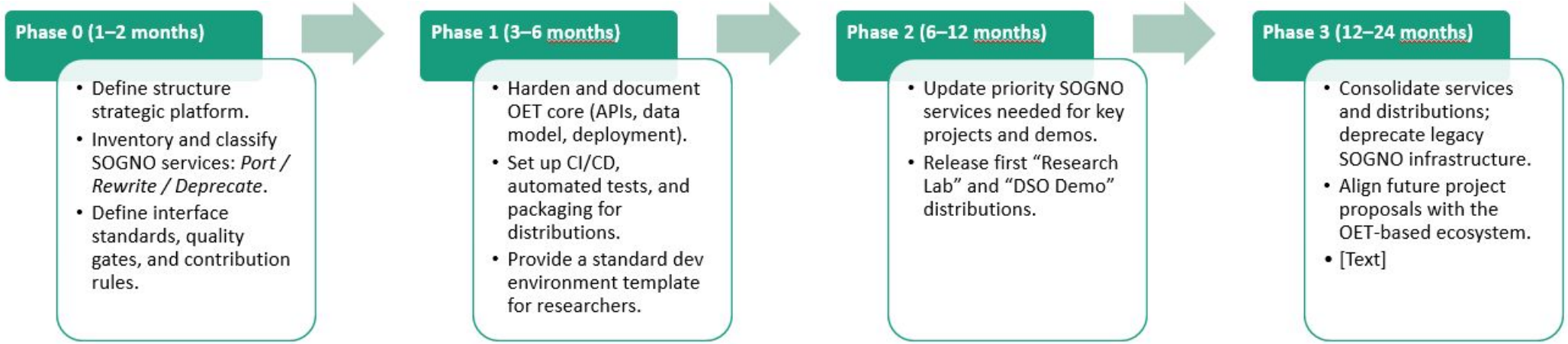
Workshops Objectives

- Understanding core blockers in current SOGNO architecture.
- Reviewing OET architecture for systematic integration and usage of the OET core and GUI.
- Identifying problems related to overlapping platform (OET & SOGNO)
  - Duplicated effort
  - Integration problems
  - Unclear Direction

Key Achievements in the past year

Workshop Results:

- Roadmap and Governance:



Governance:

- Steering group (management + lead engineers): roadmap, major decisions.
- Platform maintainers: OET core, data model, APIs, releases.
- Service owners: specific services (incl. SOGNO-derived), quality and evolution.
- Distribution maintainers: coherence, documentation, and support for each distribution.

Key Achievements in the past year

Workshop Results:

- Alignment on Architecture enhancement
- Results obtained:
  - Expectations from digitalization of SCADA Systems
  - Architecture modules and improvement potentials
  - Suggestions on Governance of Open-Source Software.
  - Established tasks and roadmap align SONGO and OET efforts toward a unified platform.

New key project: OpenAIFlex

- OpenAIFlex is a contract with the European Commission led by INESC-TEC. Fraunhofer is a sub-contractor.
- Thanks to this project some key milestones will be achieved:
  - Integration of the developments made in Rome back in the main line of the project
  - Installation of the SOGNO platform in Vienna and in other DSO in Austria
  - Replica of the experience also in one DSO in South America (negotiation in these days)
  - Creation of a dedicated API for AI-based services
  - Deployment of a first set of AI-driven services in all the active SOGNO locations including Rome

Roadmap 2026

Category	Feature/Improvement/Cooperation	Planned Quarter	Project
New Features	Grid Situational Awareness: Dynamic State Estimator as a micro-service	On Hold	
Improvements	Extend PyVolt to include DC State Estimation	Q2	
	Use case, requirements, architecture draft	Q2	InterScada OET Seguro Joint Sogno Document
	OpenSSF requirements: Best practices	Q3	SOGNO
	OKR Framework for Project management	Running with Phase 0 below	SOGNO
	Security alerts (Dependabot): Proloaf, Pymfm	2026/Q2	
	Complete Evrich REST API	To be defined	
	Set uniform Data Management	Running with Phase 1 below	
	Revise communication schema and make it uniform (DPsim + Pyvolt)	Running with Phase 2 below	
Management Proposed Roadmap	**•Phase0 (1–2 months)**		
	Define structure strategic platform.	2026/Q2	SOGNO
	Inventory and classify SOGNO services: *Port / Rewrite / Deprecate*.	2026/Q2	SOGNO
	Define interface standards, quality gates, and contribution rules.	2026/Q2	SOGNO
	•Phase 1 (3–6 months)		
	Harden and document OET core (APIs, data model, deployment).	2026/Q3	SOGNO
	Set up CI/CD, automated tests, and packaging for distributions.	2026/Q3	SOGNO
	Provide a standard dev environment template for researchers.	2026/Q3	SOGNO
	•Phase 2 (6–12 months)		
	Update priority SOGNO services needed for key projects and demos.	2026/Q4	SOGNO
	Release first “Research Lab” and “DSO Demo” distributions.	2026/Q4	SOGNO
	•Phase 3 (12–24 months)		
	Consolidate services and distributions; deprecate legacy SOGNO infrastructure.	To be defined	SOGNO
	Align future project proposals with the OET-based ecosystem.	To be defined	SOGNO

Growth Plan: Projects

- **CreteValley:** datafev will be utilized to calculate EV flexibility

- **HEDGE-IoT**
  - Utlize *ProLoaF* for demand forecasting as first step of congestion forecasting
  - Update *ProLoaF* with SOTA algorithms(Transformer, TFT)

Areas the project could use help on

- Enhance collaboration with other LF projects
  - Exploring opportunities in the field of Data Spaces (OneNet).
  - Connectors to interact with other platforms in the LFE ecosystem (EVerest).
- Growth opportunities:
  - Promoting existing SOGNO services to potential users.
  - Jointly searching funding opportunities for future developments.
- Guidance on management and quality assessment would be highly appreciated

OpenSSF Best Practices



sogno-platform

Projects that follow the best practices below can voluntarily self-certify and show that they've achieved an Open Source Security Foundation (OpenSSF) best practices badge. Show details

If this is your project, please show your badge status on your project page! The badge status looks like this: openssf best practices | in progress 79% Here is how to embed it: Show details

These are the passing level criteria. You can also view the silver or gold level criteria.

Baseline Series: Baseline Level 1 | Baseline Level 2 | Baseline Level 3

Expand panels | Show all details | Show only incomplete criteria

- Basics 13/13
- Change Control 9/9
- Reporting 8/8
- Quality 6/13
- Security 14/16
- Analysis 3/8

Status of the OpenSSF Badge

- OpenSSF's certification is discussed in our regular internal meetings
- SOGNO has several repositories and contributors to align
- Verifying non-compliant repositories and disseminating the requirements among colleagues
- We have more manpower now (RWHT+FIT), and we are refining our workflow
- The Silver Badge is our final target for the whole SOGNO platform

Feedback on working with LF Energy

- Helps to identify common elements that can then be used as a framework to expand the SOGNO suite more cleanly.
- Motivates to produce reusable, open source, and well-documented code during research projects, thus also improving the transfer of learning.
- Identification and alignment with the industry requirements thanks to the open community environment.
- Automated tools are very helpful: License scan and Insights.
- Reviewing the requirements for best practices gives us a new way of working

TAC Open Discussion